Human-Computer Interaction

5. Data Collection (2)



Last week

- Data collection method
  1. Observation
  2. Interview

Recap. Interview



A method of asking questions & listening

- Ask what you can't observe
- Hear about people's thoughts and experiences:
  - What they do
  - How they do things
  - Their opinions on current activities
  - How much they like one thing compared with another

Recap: Pros & Cons of Interview

- Pros
  - "Deep" rich data
  - Interactive
  - Open-ended
  - Exploratory
  - More economical in some cases than observing the user
- Cons
  - Time-consuming (both in conducting interviews and doing analyses)
  - Can be inconsistent and subjective
  - Amount of data can overwhelm
  - Interviewee can be biased by interviewer, or what they say might be different from what they do

Recap: Interview Questions

1. Get specific
   - Ask for tasks, roles and details of tasks
   - Ask about specific moment in the past: walk through tasks “yesterday”
   - Ask for examples!!!
   - Example: “What goes through your head when an online purchase fails?” vs. “Tell me what went through your head the last time you tried to buy something online and the purchase failed.”
2. Don’t ask questions asking frequency (How often you do X?)
3. Don’t ask how much they like things on an absolute scale.
4. Don’t ask questions which you can easily predict the answers
5. Don’t ask questions that will influence the answer
   - Example: “How anxious do you feel when an online purchase can’t be completed successfully?” vs. “Try to remember the last time an online purchase couldn’t be completed for some reason. How did you feel then?”
6. Don’t ask what they would do/like/want; new features

Recap. Structure of Interview Script

1) Introduction

   Example: "Hi Alice, thanks for coming down. My name is _____, and I'll be conducting the interview today. We're doing some market research, so some of the questions might be a little personal, and will concern your lifestyle. If you feel uncomfortable answering any of the questions, just let me know and we can skip it. I can't tell you much about what we do or the exact reason we're doing this research now, because it might affect the way you answer the questions. But I can definitely tell you after the interview is over. Do you have any questions so far? If you don't mind, I would record the audio for this interview; it will only be used internally, for the purposes of this research."

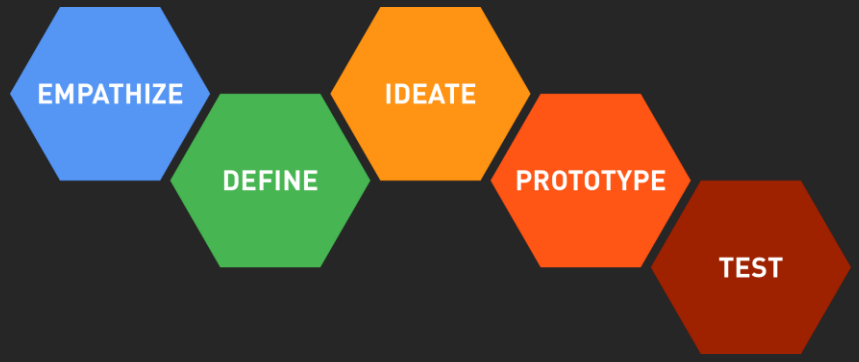
2) Warm up questions
   - Examples of warm up questions: "What's your occupation / what do you do all day?", "Can you tell me more about your hobbies?", "How often do you use the internet? etc."

3) Main body questions
   - Start with questions relating to specific incidents in the past, before moving on to speculative ones.
   - Follow a flow that makes sense to the respondents

4) Wrap up
   - Give your respondent a sense of closure.
   - Finally, thank them for their time and contribution towards your research.

Today

- Data collection method
  1. Observation
  2. Interview
  3. Survey



Define:
Collecting Data from Users

Interview Stages

1. Plan the interview
2. **Conduct the interview**

Introduction

1. Introduce yourself and any other team members present and their roles
2. Explain your purpose
3. Get consent/permission to record
    - Any risks, psychological or otherwise?
    - How will you achieve confidentiality and data security?

Build rapport

Compare these poses







Intents are important

Ask “Why?” and “How?” until you get an interesting answer or subject gets frustrated.

“Tell me more!”

Follow-up Questions

Ask for more details on anything that sounds interesting

Repeat/rephrase/summarize to make sure you understood

- Ask why
- Ask for an example
- Clarify terms
- Ask for step-by-step actions
- Make the interview to be about them, not you

Q: Mr. Moore let me ask you, what about the difference, the growing gap between those schools that have computers and those that don't?

A: That is a real prob...

Q-: It's very costly to do this isn't it?

Let him finish the first question, then ask "How costly is it to set up computers in schools?"

Ending the Interview

Leave background for last (and use a questionnaire)

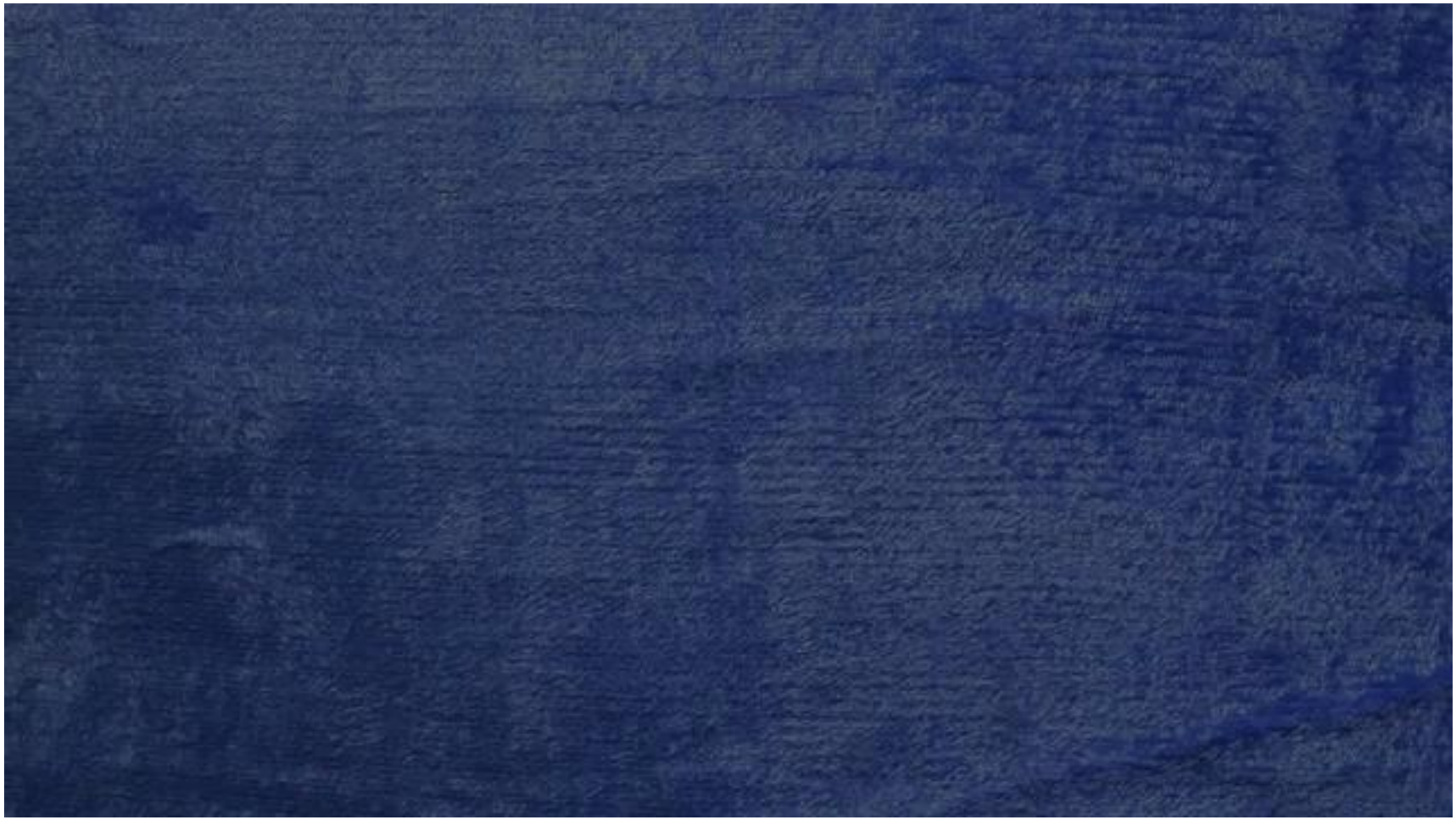
Ask:

- Any suggestions?
- Last thoughts?
- Things we did not discuss?

Finally:

- Respect, relationship building
- Get the important stuff out
- Thanks & contact details

Conduct an interview



https://www.youtube.com/watch?feature=player_detailpage&v=9t-_hYjAKww#t=486

3. Survey

Gathering precise information

- Open or Closed questions

1. How old are you? _____ years.

2. What do you like best about your job?

_____
_____
_____
_____

1. How old are you?
   (a) 12 - 15 years old
   (b) 16 - 25 years old
   (c) 26 - 35 years old
   (d) 36 - 45 years old
   (e) Over 46 years old

2. What do you like best about your job?
   (a) The people
   (b) The diversity of skills you need to do it
   (c) The pay and/or benefits
   (d) Other: _____ (write in)

3. Survey

Types of survey questions

- Mutually exclusive choices (radio buttons)
- Non-mutually exclusive choice (checkboxes)
- Ranges (overlapping, open-ended)
- Likert Scale
- Short answer fill-ins
- Comments

Question Type	Example
Choose one	Which of the following sports is your favorite? ○ Baseball ○ Basketball ○ Football ○ Hockey ○ Other _____ ○ None
Choose many	Which of the following sports do you play regularly? ☐ Baseball ☐ Basketball ☐ Football ☐ Hockey ☐ Other _____ ☐ None
Rating	Please rate your overall level of enjoyment watching your favorite sport on television. Completely Dissatisfied ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ Completely Satisfied 1 2 3 4 5 6 7 8 9 10
Ranking	Please rank these sports in order from your favorite (#1) to least favorite (#4). ___ Baseball ___ Basketball ___ Football ___ Hockey
Open-ended	Please describe why [xxx] is your favorite sport? _____ _____ _____

3. Survey

Likert Scale （李克特量表）

- Best used to measure and evaluate customer sentiment on a specific product, service or experience
- Response labeling is important!
- Common Likert Scale questions:
    1. Agreement
    2. Likelihood
    3. Satisfaction
    4. Importance

Likert Scale: Agreement

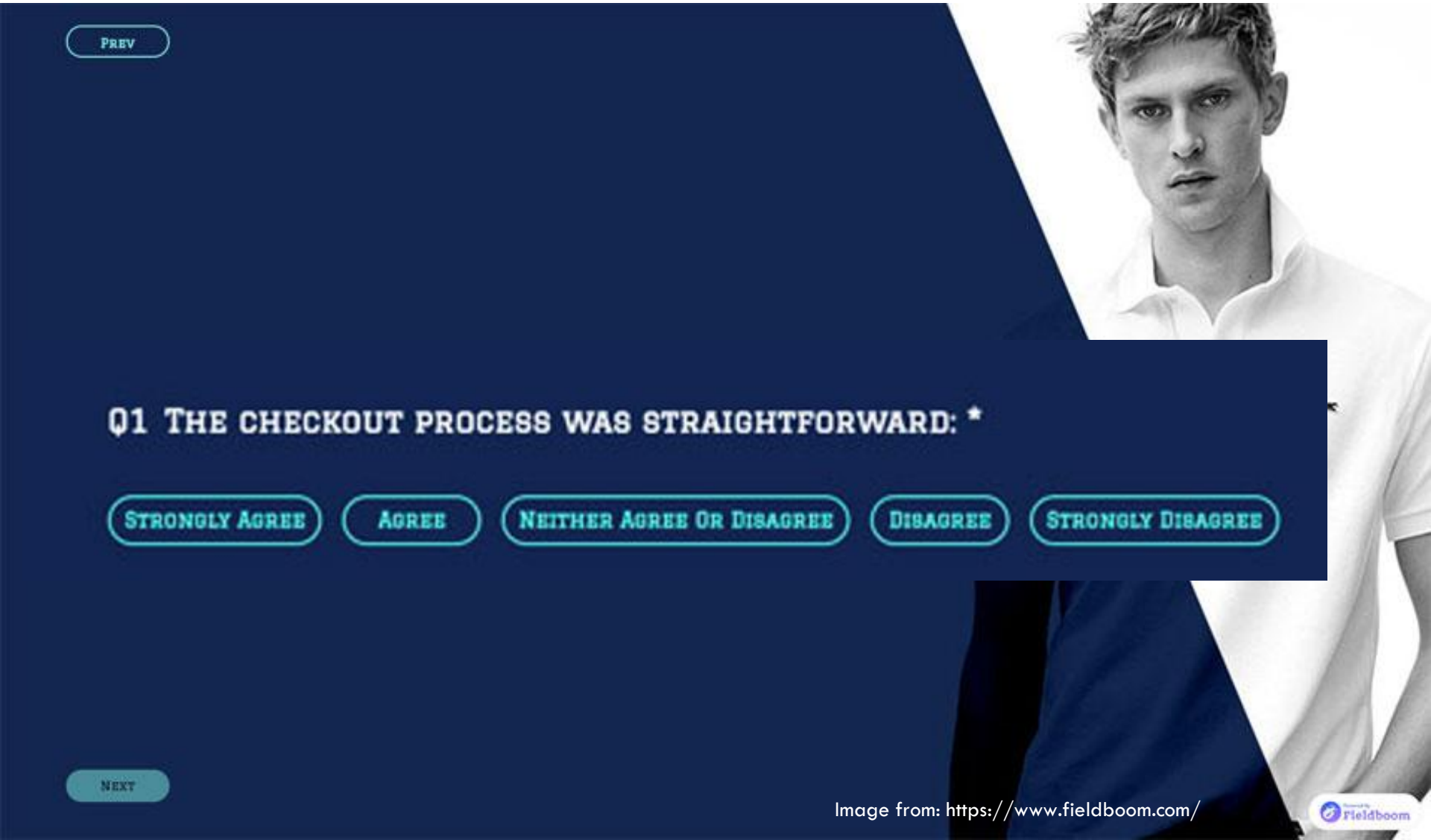


Image from: https://www.fieldboom.com/

Likert Scale: Likelihood

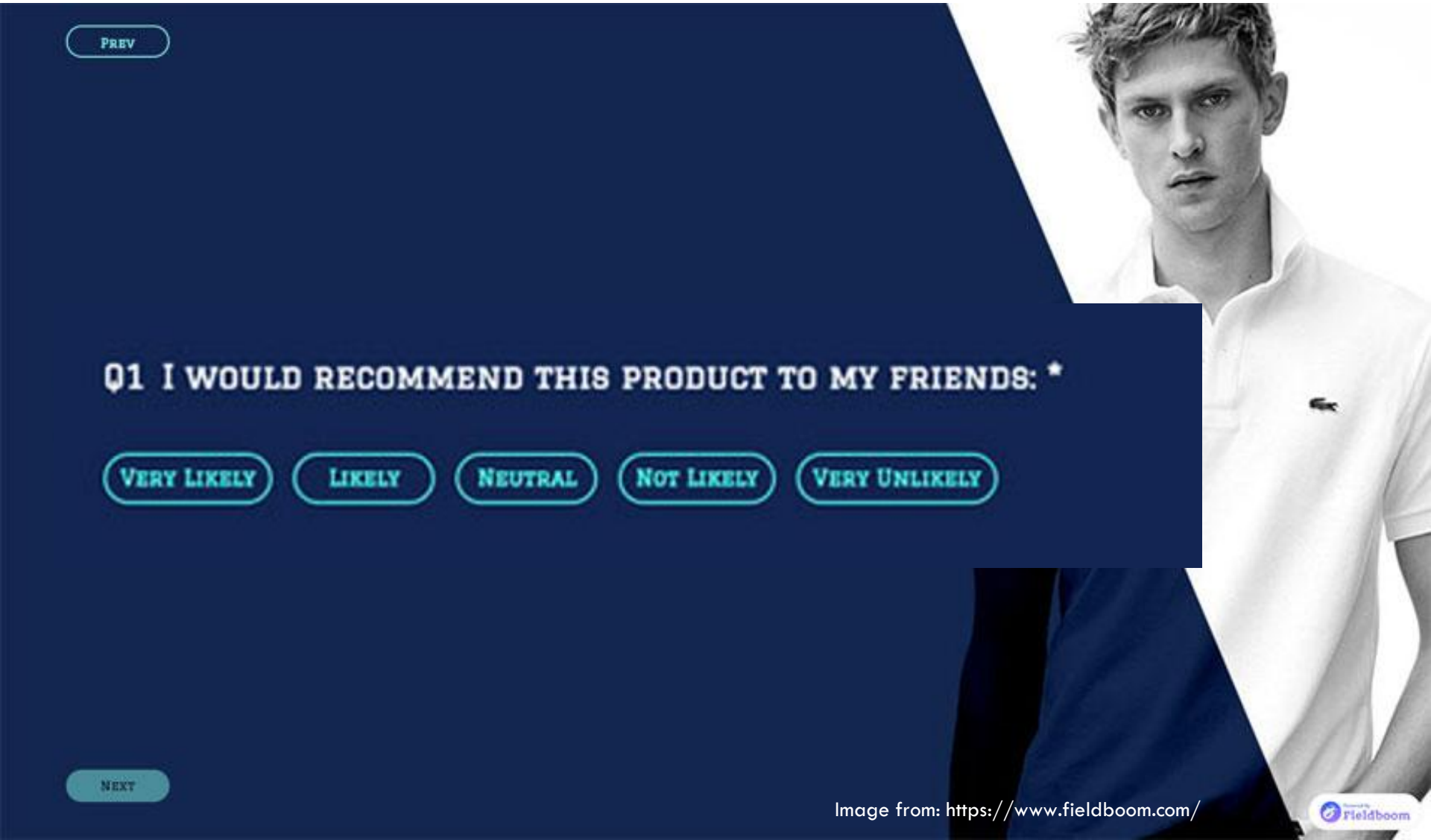


Image from: https://www.fieldboom.com/

Likert Scale: Satisfaction

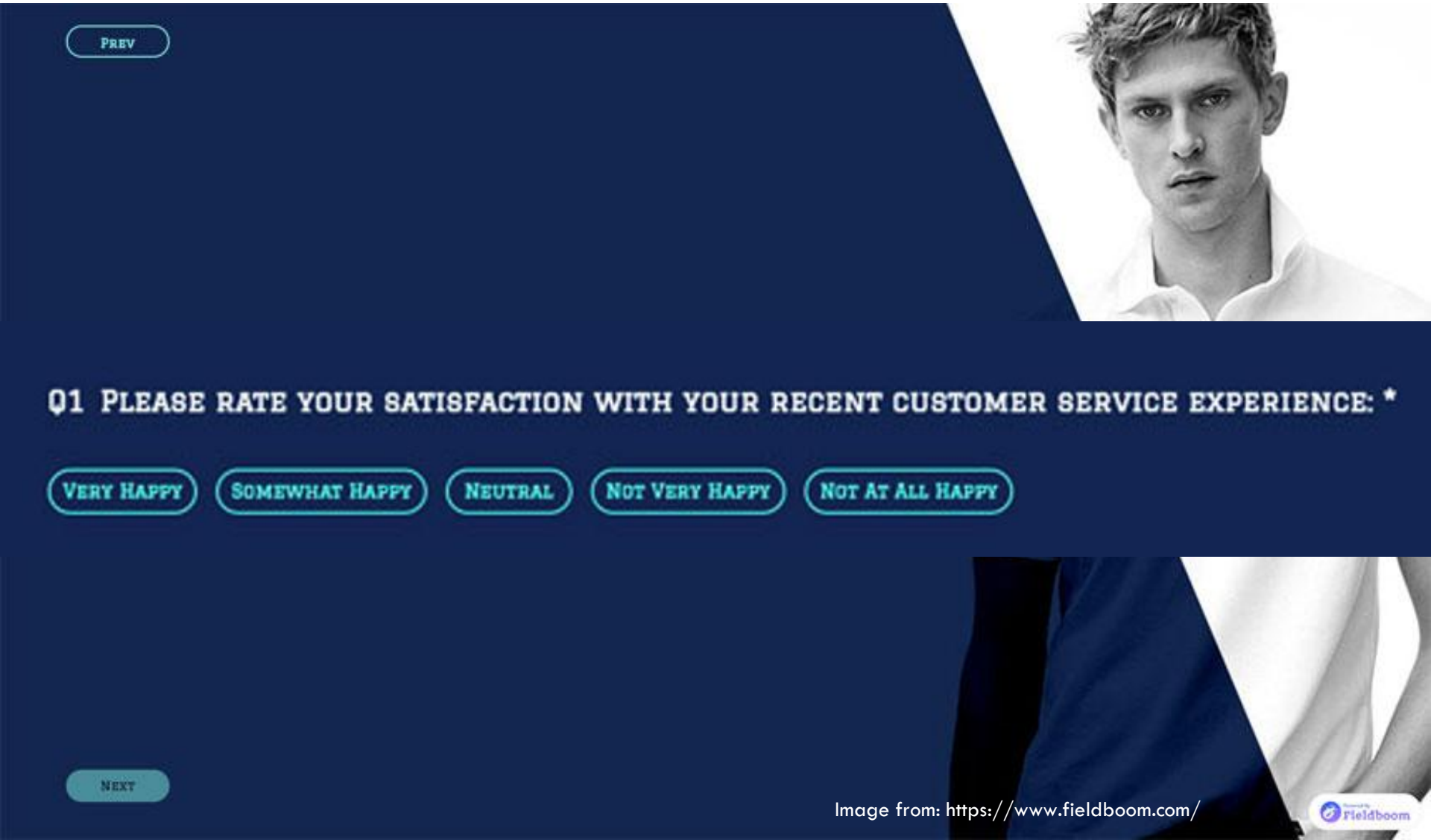


Image from: https://www.fieldboom.com/

Likert Scale: Importance



Image from: https://www.fieldboom.com/

Guidelines for surveys

1. Identify the scale and the significance of the unit
   - Use the most intuitive order
   - You can use positive or negative scales or a combination
2. Use odd numbers when you want to allow neutral responses
3. Use even numbers when you want to force a choice: positive or negative
4. Provide a not applicable (N/A) when appropriate
5. Do not use too many degrees within the scale; seven is considered a general limit
6. Test your questionnaires before use

Interview or Survey?





Interview or Survey?

Interviews

- Explore a domain
- Probe & follow-up
- Capture multiple data sources
- Rapport-building → Can ask about sensitive issues

But:

- Expensive
- Requires coding to interpret data
- Difficult to make comparisons

Surveys

- Defined sample from larger population → Generalizable
- Comparatively low cost
- Ask multiple questions efficiently
- Standardized data allows comparisons
- Quantitative results

But poor for:

- Starting research
- Determining questions to ask
- Clarifying questions for respondent
- Following up unexpected comments

主观题 10分

设置

以十一假期为主题，3分钟设计调查问卷，2分钟完成调查。

作答

Group project

P2: Planning Data Collection

Design a semi-structure interview script and a list of survey questions that you will use to collect data from potential users.

How many questions? At least 3 topic areas (themes) and 5 questions per topic for an interview; at least 15 questions for a survey. But, you need to make sure that you have enough questions to know about the users, their context, and the task. If I find important topics (based on your proposal) are missing, you will loose points.

- Do not ask the same question in an interview and a survey
- You should think deeper why you are asking Q1 in an interview, not in a survey (and vice versa)

P2: Planning Data Collection

- Turn-in: one PDF file that includes an interview script & a list of survey questions
- Format:
  - On top: Put your team name and a brief description of your project
  - Style: 12 point scale in Times New Roman, 1.5 line spacing
  - At the bottom: State who contributed to the original writing of each section and who helped improve and/or review

P2: Planning Data Collection

Rubrics

- Interview (2.5pt)
  - Structure: intro-main body-closing (1pt)
  - Quality of questions
    - Whether you ask relevant questions to your topic/problem (0.5 pt)
    - Whether you ask in-depth contextual questions (0.5pt)
    - Whether the answers to questions lead to in-depth, follow-up questions (0.5pt)
- Survey (2.5pt)
  - Whether you used proper types of questions (e.g., open ended vs. radio button) (1pt)
  - Whether you ask relevant questions to your topic/problem (1 pt)
  - Whether you ask appropriate demographic information questions (0.5pt)

P2: Planning Data Collection

Rubrics

- You will lose 50% if your submission does not follow the format
- You will lose 30% if it's a late submission (within 24 hours)
- You will lose 10% per missing an important topic
- You will lose 10% if the number of questions doesn't meet the minimum

Sample Interview Script

- Intro -

Hello user thank you for your time. I am collecting data for a research project pertaining student safety, is it alright if I use your response as part of my project? You may find some of my questions too personal, if you are uncomfortable with answering them just let me know and I will move on. Do you have any questions for me before I start?

- Warm Up -

How many classes are you taking this semester? How often do you commute from one campus to another?

Is this your first time on your own? If so how do you think you are handling it? If not what prior experiences have you had where you have been primarily responsible for your own well being?

- Main Body -

- Topic 1: Student Safety
  - Q1. Would you say that you felt prepared to go to college? Describe any new responsibilities or concerns that came as a surprise to you.

...

- Topic 3: Campus Safety and Concerns
  - Q1. Do you think that there are any specific areas (on or near campus) that could be unsafe for students?

    Follow Up: What is your overall opinion about the safety of this campus?
  - Q2. What do you think the campus security does to communicate with students in an emergency?
  - Q3. Does the campus security provide any way to connect them with students during an emergency? Do you think if it's effective?
  - Q4. Are there any safety measures that campus security takes to protect students who live on campus or off campus?
  - Q5. What do you think the campus is doing to spread safety awareness?

    Follow-up: What are your thoughts and concerns about the campus safety awareness?

- Closing -

How would you feel about a phone application that was able to alert your guardians / loved ones in the event of an emergency?

Follow Up: Do you think that you would personally use this kind of service? Why or why not?

Thank you for your responses!

Sample Survey

- Q3. What is your marital status? (Radio buttons - Single, Married, Divorced, Other [with comments])
  - Single
  - Married
  - Divorced
  - Other (Explain) _____
- Q4. What is your employment status? (Radio buttons - Unemployed, Salaried Worker, Wage Worker, Self-Employed, Other [with comments])
  - Unemployed
  - Salaried Worker
  - Wage Worker
  - Self-Employed
  - Other (Explain) _____
- Q5. How frequently do you contact your doctor's office to schedule an appointment? (Likert Scale - Daily, Weekly, Monthly, Annually)

  Daily Weekly Monthly Annually
- Q6. How do you typically contact your doctor's office? (Radio Buttons - Email, Phone Call, Text Message, Proprietary Application [with comments])
  - Email
  - Phone Call
  - Text Message
  - Proprietary Application (Explain) _____
- Q7. How responsive is your doctor's office in responding to your messages/voicemails/etc.? (Likert Scale - Very Unresponsive, Unresponsive, Average Responsive, Responsive, Very Responsive)

Very Unresponsive Unresponsive Neither Unresponsive/Responsive Responsive Very Responsive

- Q8. Who do you most often speak to when making a doctor's appointment? (Checkboxes - Doctor, Nurse, Nurse's assistant, Receptionist, Other [with comments])
  - ❑ Doctor
  - ❑ Nurse
  - ❑ Nurse's Assistant
  - ❑ Receptionist
  - ❑ Other (Explain) ____
- Q9. How many different apps do you typically use on your phone? (Range - 1 _ _ _ _ _ 10+)

  1 2 3 4 5 6 7 8 9 10 10+
- Q10. How comfortable are you with using technology in your everyday life? (Likert Scale - Very Uncomfortable, Uncomfortable, Neither Uncomfortable nor Comfortable, Comfortable, Very Comfortable)

Very Uncomfortable Uncomfortable Neither Uncomfortable/Comfortable Comfortable Very Comfortable

- Q11. Do you use your phone to store sensitive information? (Passwords, bills, bank statements, etc) (Open-ended - Explanation of what kind of sensitive info, parameters for storing certain info)

  _____
  _____
  _____
- Q12. How often do you use video chat software? (Google Hangouts, Skype) (Likert Scale - Daily, Weekly, Monthly, Annually, Sparsely)

  Daily Weekly Monthly Annually Sparsely
- Q13. Do you use your phone to track your health? (Fitness monitoring, diet tracking, etc) (Multiple Choice/Open-ended - If yes, Explanation of what kind of data is tracked, how it is tracked, etc.)